

Arjun Garg

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TECHNICAL SKILLS

Design & Fabrication: CAD modelling, 3D printing

Programming: Python, C, C++, JavaScript

Hardware: circuit board prototyping, mechanical/electrical troubleshooting

Certifications: National Lifeguard, Standard First Aid & CPR-C

EXPERIENCE

Team Engineer

Summer 2025

SHAD Program – Memorial University

St. John's, NL

- Calculated the RPM band needed to keep a three-wheeled prototype vehicle's motor efficient at downtown driving speeds, then built a working mini-prototype using CAD models, custom circuit boards, and 3D-printed parts
- Diagnosed a gearbox failure under vibration, traced it to a flexing housing misaligning the gears, redesigned the part, and learned KiCad to clean up the prototype's PCB layout – prototype won the Best Prototype award

General Member

2024 – 2025

FIRST Robotics Team 1325, Inverse Paradox

Mississauga, ON

- Worked hands-on with the team's drive shaft, diagnosing intermittent robot failures across mechanical, electrical, and software subsystems
- Traced a recurring in-match shutdown to a mismatch between commanded controller output and what the electrical system could deliver, rather than a mechanical fault; team reached the Ontario Provincial Championship District Finals (Quality Award)

Swim Instructor & Lifeguard

2025 – 2026

LIFETIME Fitness

Mississauga, ON

- National Lifeguard certified; teach swimming lessons twice weekly, designing lesson plans that are safe, age-appropriate, and engaging for each student's skill level

LEADERSHIP & ACTIVITIES

Founder & President

2016 – Present

CandyForKids.org

Mississauga, ON

- Founded and grew a charity redistributing excess Halloween candy, scaling from door-to-door drives to a permanent GTA donation-box network and pickup service; 2,000+ candy bags donated to date
- Led a 6-person executive team through a last-minute logistics crisis when a partner food bank withdrew with two days' notice – set recipient criteria, ran a time-boxed emergency meeting, and coordinated a new international donation; now runs the charity part-time while at Waterloo

General Executive

2024 – 2026

Peer Mentoring, John Fraser Secondary School

Mississauga, ON

- Trained senior students to support incoming students and Grade 9s; spoke at school events and worked one-on-one with mentees on issues ranging from social anxiety to accessing school and volunteer resources

Competitor

2022 – 2026

DECA

Mississauga, ON

- Qualified for DECA's International Career Development Conference as part of the Ontario delegation;
- Top 20 Overall Finalist and Top 20 Oral Event performer at Provincials (2026)

EDUCATION

University of Waterloo

Waterloo, ON

B.A.Sc. in Mechatronics Engineering, Stream 8 (Co-op)

Sept. 2026 – Expected 2031

John Fraser Secondary School

Mississauga, ON

Advanced Placement Program

2022 – 2026

- AP Precalculus: 5, AP Physics 1: 5, AP Chemistry: 4, AP Computer Science Principles: 4
- Graduated the Spirit of Math enrichment program with Honours (2025)